6.7 Dedicated Multipliers in FPGAs

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- Dedicated multipliers:
  - will be more ***area efficient*** and will be ***faster*** than multipliers realized using logic blocks.
- Many commercial FPGAs provide dedicated multipliers.
  - E.g.s: Xilinx Virtex-4/Spartan-3 and Altera Stratix/Cyclone FPGAs contain 18×18 multipliers.

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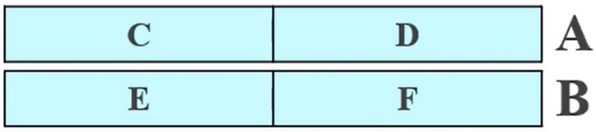
- When multiplication of numbers larger than 18 bits is required, several of the ***dedicated built-in multipliers*** can be put together.
  - E.g.: If A and B are 32 bits, and C, D, E, and F are the 16 bit components of A and B

$$A = C \times 2^{16} + D$$
$$B = E \times 2^{16} + F$$



$$\Rightarrow AB = \mathbf{CE} \times 2^{32} + (\mathbf{DE} + \mathbf{CF}) \times 2^{16} + \mathbf{DF}$$

$\Rightarrow$ ***Four multipliers*** are required to generate the partial products CE, DE, CF, and DF, and ***several adders*** are required to add the partial products.

Synthesis Tools

$A = C \times 2^{16} + D$

$B = E \times 2^{16} + F$

$\Rightarrow AB = \mathbf{CE} \times 2^{32} + (\mathbf{DE} + \mathbf{CF}) \times 2^{16} + \mathbf{DF}$

- Synthesis tools are capable of inferring ***dedicated multipliers*** on FPGAs that provide them.
  - E.g.: Verilog code that infers dedicated multipliers

```
module multiplier (A, B, C);
  input [31:0]   A;
  input [31:0]   B;
  output [63:0]  C;
  wire [63:0]    C;
  assign C = A * B ;
endmodule
```

  - When code is synthesized for Xilinx Spartan devices using Xilinx ISE tools, the synthesis tool infers ***4 dedicated 18 × 18 multipliers***, and ***several logic blocks*** are used in ***addition*** to the 4 multipliers.